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A mixed-integer quadratically-constrained programming model for the distribution system expansion planning

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ABSTRACT

This paper presents a mixed-integer quadratically-constrained programming (MIQCP) model to solve the distribution system expansion planning (DSEP) problem. The DSEP model considers the construction/reinforcement of substations, the construction/reconductoring of circuits, the allocation of fixed capacitors banks and the radial topology modification. As the DSEP problem is a very complex mixed-integer non-linear programming problem, it is convenient to reformulate it like a MIQCP problem; it is demonstrated that the proposed formulation represents the steady-state operation of a radial distribution system. The proposed MIQCP model is a convex formulation, which allows to find the optimal solution using optimization solvers. Test systems of 23 and 54 nodes and one real distribution system of 136 nodes were used to show the efficiency of the proposed model in comparison with other DSEP models available in the specialized literature.

1. Introduction

The distribution system expansion planning (DSEP) aims to find the best proposal of expansion for the system that minimizes the investment and operation costs, while satisfying operational constraints, such as voltage magnitude limits at buses, current flow magnitude limits in circuits, maximum apparent power in substations and a radial configuration of the system, for a planning horizon in which demand is known. Thus, the DSEP can add the following to system: (a) new substations and/or repower existing ones; (b) change the conductors of existing circuits and/or build new circuits in branch candidates; and (c) allocate of capacitors and/or voltage regulators, [1].

According to the planning horizon, the DSEP problem can be classified as a short-range (1 to 4 years) or long-range (5 to 20 years) problem [2]. According to the model, one can have a static problem, in which only one stage (planning horizon) is considered, or a dynamic problem (multistage problem), in which several planning horizons are considered in the same problem [3]. The latter problem is not addressed in the present paper. Mathematically, the DSEP problem is modeled as a mixed-integer non-linear programming (MINLP) problem and has been solved using heuristic algorithms, meta-heuristic techniques and also classical optimization techniques such as mixed-integer linear programming (MILP), MINLP and quadratic programming (QP) [4].

Reference [5] presents a MILP model for the integral planning of primary-secondary distribution systems. In [6], a MILP model for the multi-stage DSEP problem is presented considering the available capacities of distributed generators in the system. Mixed-integer programming models determine the optimal location of distribution substation and feeder expansion are developed in [7,8].

In [9,10], QP is used to model the DSEP problem. A heuristic constructive algorithm for the DSEP problem, that approximates the real power losses using a square function is proposed in [9]. The algorithm relaxes the integrality of the decision variables and solves the resultant QP problem to determine the variables that can be rounded. In [10], a two-phases iterative technique is used, that determines the optimal substation sites in the first phase, while the second phase selects the configuration of the network; the integrality of the variables is relaxed allowing to solve a QP problem, and, along with its solution, integer constraints are imposed using heuristic techniques.

In [11], a constructive heuristic algorithm (CHA) is presented to solve the DSEP problem. In this work, the CHA uses a sensitivity index obtained by the solution of a non-linear programming model. Additionally, a local improvement phase is implemented. [12] presents a conic programming model for the DSEP problem. In that paper, two formulations are analyzed: the single-circuit and the parallel equivalent circuit. Additionally, constraints are proposed to eliminate loops with the aim of obtaining a tight formulation

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and reducing the computational effort necessary to solve the DSEP problem. A MINLP model for the DSEP problem is proposed in [13], formulating generalized radiality constraints considering multiple substations, distributed generator and transfer nodes (nodes without load). The presented model is solved using an algorithm based on the non-linear branch and bound technique, and the non-linear programming problems are solved using a commercial solver. Heuristic algorithms have been successfully applied to solve the DSEP problem [14–17]. An algorithm based on the branch exchange technique was developed in [16], while in [17] this technique is improved upon by using a heuristic procedure to obtain better solutions for the network by adding trans-shipment nodes. Several works applied meta-heuristics in the solution of the DSEP problem [18–22]. In [18,19] genetic algorithms are used. A method based on ant colony search was developed in [20], while [21,22] used simulated annealing.

The reliability of the system has been considered using multi-objective approaches for the DSEP problem, as can be seen in [23][26]. In [23,24], evolutionary algorithms were used to solve the DSEP problem, taking into account monetary costs and the system failure index. In [25], multi-objective algorithms, such as NSGA and SPEA, were applied to solve the design of distribution systems. Ref. [26] presented a method to obtain a Pareto front for the multi-objective DSEP problem, employing a dynamic programming approach.

In this paper, a mixed-integer quadratically-constrained programming (MIQCP) model to solve the DSEP problem is proposed. The DSEP model considers the construction/reinforcement of substations, the construction/reconductoring of circuits, the allocation of capacitors banks and the radial topology modification of the system. In contrast with other works on the DSEP problem, the proposed model considers the allocation of fixed capacitors banks in the DSEP problem. As the DSEP problem is a very complex MINLP problem, it is convenient to reformulate it like a MIQCP problem; it is therefore demonstrated that the proposed formulation represents the steady-state operation of a radial distribution system. The MIQCP model is a convex formulation that allows to find the optimal solution of the problem using optimization solvers. In contrast with other MIQCP formulations for the DSEP problem [12], the proposed model uses the original variables of the power flow of the distribution system (voltage magnitudes, current flow magnitudes and power flows). Test systems of 23 and 54 nodes and one real distribution system of 136 nodes were used to show the efficiency of the proposed model in comparison with other DSEP models available in the specialized literature.

The main contributions of this paper are as follows:

1. A novel quadratically-constrained programming model to obtain the steady-state operation of a radial distribution system.
2. A MIQCP formulation for the DSEP problem with the following benefits: (a) a robust mathematical model that finds the same solution of the MINLP model; (b) an efficient computational behavior with conventional MIQCP solvers; (c) convergence to optimality is guaranteed using classical optimization techniques.
3. The DSEP model considers the construction/reinforcement of substations, the construction/reconductoring of circuits, the allocation of capacitors banks and the radial topology modification.

2. The distribution system expansion planning problem

2.1. Steady-state operation of a radial distribution system

The equations that represent the steady-state operation of a radial distribution system are presented in (1)–(4). Note that (1)–(3) are linear, while (4) is non-linear, containing square terms and the product of two variables. These equations are frequently used in the load flow sweep method of radial distribution systems [27,28], and can be used to formulate an MINLP model for the DSEP problem.

$$\sum_{ki\in\Omega_l}\widehat{P}_{ki}-\sum_{ij\in\Omega_l}\left(\widehat{P}_{ij}+R_{ij}\widehat{I}^{sqr}_{ij}\right)+P^S_i=P^D_i\quad\forall i\in\Omega_b \tag{1}$$

$$\sum_{ki\in\Omega_l}\widehat{Q}_{ki}-\sum_{ij\in\Omega_l}\left(\widehat{Q}_{ij}+X_{ij}\widehat{I}^{sqr}_{ij}\right)+Q^S_i=Q^D_i\quad\forall i\in\Omega_b \tag{2}$$

$$V^{sqr}_i-V^{sqr}_j=2(R_{ij}\widehat{P}_{ij}+X_{ij}\widehat{Q}_{ij})+Z^2_{ij}\widehat{I}^{sqr}_{ij}\quad\forall ij\in\Omega_l \tag{3}$$

$$V^{sqr}_j\widehat{I}^{sqr}_{ij}=\widehat{P}^2_{ij}+\widehat{Q}^2_{ij}\quad\forall ij\in\Omega_l \tag{4}$$

Considering the following four assumptions, the steady-state operation of a radial distribution system can also be obtained using a quadratically-constrained programming problem, as shown in (5): (a) the real power losses are being minimized in an objective function, which implies that the current flow magnitudes in the branches are also minimized, supposing $R_{ij}>0$; (b) a radial operation of the distribution system; (c) V^{sqr}_j and $\widehat{I}^{sqr}_{ij}$ are non-negative variables; and (d) the Lagrange multipliers of the second-order cone constraints (5.b) are greater than zero.

$$\left.\begin{array}{ll}\min \sum_{ij\in\Omega_l}R_{ij}\widehat{I}^{sqr}_{ij} & \text{(a)}\\ \text{subject to} & \\ \text{constraints (1)–(3).} & \\ V^{sqr}_j\widehat{I}^{sqr}_{ij}\geqslant\widehat{P}^2_{ij}+\widehat{Q}^2_{ij}\quad\forall ij\in\Omega_l & \text{(b)}\end{array}\right\} \tag{5}$$

Note that (4) was re-written like a second-order cone constraint (5.b), that belongs to the set of quadratic constraints [29]. The problem (5) is convex, which makes it possible to find the optimal solution using optimization solvers [30]. In the Appendix section it is demonstrated that, in the solution point of (5), the constraint (5.b) is active and is equivalent to (4). Therefore, solving (5) is equal to solving (1)–(4), thus the quadratically-constrained programming problem (5) effectively represents the steady-state operation of a radial distribution system. In the next subsection this formulation will be used to model the DSEP problem like a MIQCP.

2.2. Mixed-integer quadratically-constrained programming model for the DSEP problem

The DSEP problem is modeled as an MIQCP problem as shown in (6)–(35). Since the proposed model for the DSEP problem considers a static formulation, the solution of the MIQCP model establishes the investments that must to be made in the system, in order to meet future demand and satisfy operational constraints.

$$\begin{aligned}\min\quad&\sum_{i\in\Omega_s}(\kappa_s c^s_i w_i+\alpha f(\tau_s,\lambda)\phi_s c^v_i S g^{sqr}_i)\\&+\sum_{ij\in\Omega_l}\sum_{a\in\Omega_a}\left(\kappa_l c^f_{ij,a}z_{ij,a}+\alpha f(\tau_l,\lambda)\phi_l c^l R_a I^{sqr}_{ij,a}\right)l_{ij}\\&+\sum_{i\in\Omega_b}(c^{fx}q_i+c^{un}n^{cp}_i)\end{aligned} \tag{6}$$

subject to

$$\sum_{ki\in\Omega_l}\sum_{a\in\Omega_a}P_{ki,a}-\sum_{ij\in\Omega_l}\sum_{a\in\Omega_a}\left(P_{ij,a}+R_a l_{ij}I^{sqr}_{ij,a}\right)+P^S_i=P^D_i\quad\forall i\in\Omega_b \tag{7}$$

$$\sum_{ki\in\Omega_l}\sum_{a\in\Omega_a}Q_{ki,a}-\sum_{ij\in\Omega_l}\sum_{a\in\Omega_a}\left(Q_{ij,a}+X_a l_{ij}I^{sqr}_{ij,a}\right)+Q^S_i+Q^{cp}n^{cp}_i=Q^D_i\quad\forall i\in\Omega_b \tag{8}$$

$$V^{sqr}_i-V^{sqr}_j=\sum_{a\in\Omega_a}\left[2(R_aP_{ij,a}+X_aQ_{ij,a})l_{ij}+Z^2_a l^2_{ij}I^{sqr}_{ij,a}\right]+b_{ij}\quad\forall ij\in\Omega_l \tag{9}$$

$$V_j^{sqr}\widehat{I}_{ij}^{sqr} \geqslant \widehat{P}_{ij}^2 + \widehat{Q}_{ij}^2 \quad \forall ij \in \Omega_l \tag{10}$$

$$\widehat{I}_{ij}^{sqr} = \sum_{a\in\Omega_a} I_{ij,a}^{sqr} \quad \forall ij \in \Omega_l \tag{11}$$

$$\widehat{P}_{ij} = \sum_{a\in\Omega_a} P_{ij,a} \quad \forall ij \in \Omega_l \tag{12}$$

$$\widehat{Q}_{ij} = \sum_{a\in\Omega_a} Q_{ij,a} \quad \forall ij \in \Omega_l \tag{13}$$

$$\underline{V}^2 \leqslant V_i^{sqr} \leqslant \overline{V}^2 \quad \forall i \in \Omega_b \tag{14}$$

$$0 \leqslant I_{ij,a}^{sqr} \leqslant \overline{I}_a^2\left(y_{ij}^+ + y_{ij}^-\right) \quad \forall ij \in \Omega_l, \forall a \in \Omega_a \tag{15}$$

$$\sum_{a\in\Omega_a} z_{ij,a} = y_{ij}^+ + y_{ij}^- \quad \forall ij \in \Omega_l \tag{16}$$

$$y_{ij}^+ + y_{ij}^- \leqslant 1 \quad \forall ij \in \Omega_l \tag{17}$$

$$\sum_{ij\in\Omega_l}\left(y_{ij}^+ + y_{ij}^-\right) = |\Omega_b| - |\Omega_s| \tag{18}$$

$$\sum_{ij\in\Omega_l} y_{ij}^- + \sum_{ki\in\Omega_l} y_{ki}^+ \geqslant 1 \quad \forall i \in \Omega_b - \Omega_s, P_i^D \geqslant 0 \tag{19}$$

$$|b_{ij}| \leqslant \overline{b}(1 - y_{ij}^+ - y_{ij}^-) \quad \forall ij \in \Omega_l \tag{20}$$

$$P_{ij,a} \leqslant \overline{VI}_a y_{ij}^+ \quad \forall ij \in \Omega_l, \forall a \in \Omega_a \tag{21}$$

$$P_{ij,a} \geqslant -\overline{VI}_a y_{ij}^- \quad \forall ij \in \Omega_l, \forall a \in \Omega_a \tag{22}$$

$$|Q_{ij,a}| \leqslant \overline{VI}_a\left(y_{ij}^+ + y_{ij}^-\right) \quad \forall ij \in \Omega_l, \forall a \in \Omega_a \tag{23}$$

$$I_{ij,a}^{sqr} \leqslant \overline{I}_a^2 z_{ij,a} \quad \forall ij \in \Omega_l, \forall a \in \Omega_a \tag{24}$$

$$|P_{ij,a}| \leqslant \overline{VI}_a z_{ij,a} \quad \forall ij \in \Omega_l, \forall a \in \Omega_a \tag{25}$$

$$|Q_{ij,a}| \leqslant \overline{VI}_a z_{ij,a} \quad \forall ij \in \Omega_l, \forall a \in \Omega_a \tag{26}$$

$$Sg_i^{sqr} \geqslant Pg_i^2 + Qg_i^2 \quad \forall i \in \Omega_s \tag{27}$$

$$Sg_i^{sqr} \leqslant Sg_i^{o2} + (2Sg_i^o Sg_{i'} + Sg_{i'}^2)w_i \quad \forall i \in \Omega_s \tag{28}$$

$$0 \leqslant n_i^{cp} \leqslant \overline{n_b^{cp}}\, q_i \quad \forall i \in \Omega_b \tag{29}$$

$$\sum_{i\in\Omega_b} q_i \leqslant \overline{n^{cp}} \quad \forall i \in \Omega_b \tag{30}$$

$$q_i \in \{0,1\} \quad \forall i \in \Omega_b \tag{31}$$

$$n_i^{cp} \in \mathbb{Z}_{\geqslant 0} \quad \forall i \in \Omega_b \tag{32}$$

$$w_i \in \{0,1\} \quad \forall i \in \Omega_s \tag{33}$$

$$y_{ij}^+, y_{ij}^- \in \{0,1\} \quad \forall ij \in \Omega_l \tag{34}$$

$$z_{ij,a} \in \{0,1\} \quad \forall ij \in \Omega_l, \forall a \in \Omega_a \tag{35}$$

The objective function (6) stands for the annualized investment and operation costs based on [11]. The first part represents the associated costs of the substations (investment and operation costs), while the second part corresponds to the associated cost of the circuits (construction/reconductoring and real power losses costs). The third part of the objective function represents the cost for fixed capacitors banks, considering the cost of their installation and the cost of each standard capacitor unit. The function $f(\tau,\lambda) = [1-(1+\tau)^{-\lambda}]/\tau$ facilitates the calculation of the present value of an annualized cost (substation operation cost or real power losses cost) in terms of the interest rate τ and the number of years of the planning period λ.

Eqs. (7) and (8) represent the real and reactive power balance in node i, respectively. Eq. (9) serves to calculate the voltage drop in circuit ij. Constraint (10) is a quadratic constraint that established the relationship between the real and reactive power flow of circuit ij, the square of the voltage magnitude (at the end of the circuit) and the square of the current flow magnitude of circuit ij; at the operation point of the model's solution, this constraint is active and therefore is equivalent to (4), as shown in Section 2.1. As the real and reactive power flows and current magnitude flow are associated with each selection of conductor type a ($I_{ij,a}^{sqr}, P_{ij,a}$ and $Q_{ij,a}$), this constraint is written in terms of the square of the total current flow magnitude and the total real and reactive power flows of the circuit ij, which are calculated using (11)–(13).

The limits for the voltage magnitude are established by (14). Constraint (15) sets the limits for the current flow magnitude of the circuit ij related to each conductor type a and its operation state (connected or disconnected). Two binary variables are used to represent the operation state of the circuit ij, with the aim of improving the performance of the solution for the DSEP problem, as was proposed for the distribution system reconfiguration problem in [32].

Eq. (16) allows the selection of one and only one conductor type for circuit ij if it is connected, while (17) allows only one real power flow direction in a circuit ij. Constraint (18), combined with (7) and (8), is used to obtain a radial topology for the DSEP problem, as shown in [13]. Constraint (19) represents the condition that each node with a demand must be connected and fed by at least one circuit. This constraint is not necessary to define the set of feasible solutions, but is included in the model in order to decrease the computational effort required for its solution.

Constraint (20) limits the variable b_{ij} in terms of the operational state of the circuit ij. That is, if the circuit is connected, then b_{ij} is 0, otherwise b_{ij} is limited by $\overline{b}$, which is chosen according to the maximum voltage drop. The real power flow in the circuit ij with conductor type a is defined by the forward and backward directions, as is shown in (21) and (22). Constraints (21)–(23) set limits for the real and reactive power flows in the circuit ij, depending on its operational state. Constraints (24)–(26) set the limits for the current flow magnitude, the active and the reactive power flow of the circuit ij related to each conductor type a.

The square of the apparent power supplied by each substation is calculated using (27) and is limited by (28). Note that (27) is a quadratic constraint, and, as the operation cost of the substations are minimized in (6), the square of the apparent power (in the optimal solution) must be equal to the sum of the square of the real and reactive power supplied by the substation.

The number of standard capacitor units installed in one node of the system is limited by (29), while (30) limits the number of capacitors installed in the system. Eqs. (31) and (32) represent the binary and integer nature of the variables for the allocation of a fixed capacitor q_i and the number of standard capacitor units installed at a node n_i^{cp}, respectively. Eqs. (33)–(35) represent the binary nature of the variables for the construction/reinforcement of substations, the state and the construction/reconductoring of the circuits, respectively. The binary investment variables q_i, n_i^{cp}, w_i and $z_{ij,a}$ are the decision variables, and a feasible operation solution for the distribution system depends on their value. The remaining variables represent the operating state of a feasible solution. For a feasible investment proposal, defined through specified values of q_i, n_i^{cp}, w_i and $z_{ij,a}$, several feasible operation states are possible.

Constraints (10)–(27) are quadratic constraints, which implies that the optimization model (6)–(35) is a mixed-integer quadratically-constrained programming model. This kind of problem can be solved efficiently by using optimization software, with the advantage that the optimality of its solution can be guaranteed due to the convexity of the formulation. The incorporation of (10) and (27) into the proposed model leads to a more tractable formulation rather than to solving the related MINLP problem directly, which is harder to solve; another alternative is to transform the model into a MILP problem using linearizations or approximations, which introduces additional variables and makes the model an approximation of the original formulation. Taking the aforementioned considerations and the results presented in Section 3 into account, the use of a MIQCP formulation for the DSEP problem is convenient.

3. Tests and results

The proposed model was tested using two test systems with 23 and 54 nodes respectively and one real distribution system with 136 nodes. The data for the distribution systems are the same as used in [11]. Additionally, for the 23-node and the 54-node test systems, the DSEP problem was solved considering allocation of fixed capacitor banks. In those cases, it was supposed that the standard capacitor unit had a value of 60 KVAr with an annualized cost of US$200 and the maximum number of capacitors that can be added to the system is 3.

It should be noted that the proposed model can be adapted to represent other optimization problems in distribution systems, such as the optimal capacitor allocation problem. In order to show that application, the proposed model was used to solve the capacitors allocation problem for a test system of 69 nodes and a real system of 136 nodes. Aiming to compare the results with previous works, the proposed model was adapted to consider switched capacitors and load levels, according to [31]. For this test, the maximum voltage magnitude and the voltage magnitude of the substation is 1.00 pu; the energy cost is 0.06US$/kWh for all load levels. The installation cost of the fixed and switched capacitors is equal to US$1000, the unit cost of each standard capacitor unit is equal to US$900 and the reactive power of each standard capacitor unit is equal to 300 kVAr. Finally, the maximum number of standard capacitor units that can be installed in a node of the system is equal to 4.

The proposed model was implemented in AMPL [33] and solved with CPLEX [30] (called with the optimality gap option equals to 0%), using a workstation with an Intel *i*7 870 processor.

A summary of the results is shown in Table 1 for the DSEP problem and in Table 2 for the capacitor allocation problem. Tables 3 and 4 show a comparison between the solutions found by the proposed method and the solutions reported by other works in the specialized literature. For all of the tests, the solution was found achieving an optimality gap of 0%, which indicates that the solution is, in fact, the optimal one. The time necessary to solve the model in each test is shown in Tables 1 and 2. It must be considered that the reported time needed for solving the model is the total time required for the solver to completely explore the Branch & Bound tree and guarantee that the solution found is the optimal one; usually the time needed to find the optimal solution is far lower than the total time.

3.1. 23-node Distribution system

The 23-node distribution system has a nominal voltage of 34.5-kV, feeds 21 load nodes and supplies 7.04-MVA. The circuits can be built using two types of conductors, with costs of 10kUS$ and 20kUS$, respectively. The minimum voltage magnitude is equal to 0.97pu, the cost of energy losses is 0.05US$/kWh, the loss factor equals 0.35, the interest rate is 0.1, the planning period extends to 20 years and the operation cost of the substations is 0US$/kVAh^2.

Table 1
Results for the test systems.

System	Time (s)	Power losses (kW)	Investment (US$)		Total cost (US$)
			Circuits	S/Es	
23-node	3	13.068	153913	0	170969
23-node + cap	2	12.650	153913	0	170824
54-node	140	0.909	40544	440000	3328955
Mod. 54-node	40	2.628	40416	280000	327277
Mod. 54-node + cap	13	2.511	39920	180000	227074
136-node	1028	259.184	5920	0	5457488

The optimization process found a solution of US$170969 in a time of 3 s. This solution is the same as the one presented in [12] and is better than the one presented in [11], which came out to a cost of US$172110. The expansion plan proposes the construction of all circuits using the first conductor type, except the circuit 1-10 which was built with the second conductor type. Additionally, circuits 3-8, 3-16, 4-6, 4-8, 4-9, 5-14, 6-16, 11-22, 12-15, 13-15, 15-21, 16-22 and 19-20 were not built. Solving the DSEP problem considering allocation of fixed capacitor banks, a solution with objective function of US$170824 was found, which allocates capacitors in nodes 3 and 9 with an investment cost of US$400. This solution is slightly lower than the one mentioned above, but both of them have the same final topology. The expansion plan for the 23-node distribution system is shown in Fig. 1.

3.2. 54-node Distribution system

The 54-node distribution system has a nominal voltage of 13.5-kV, feeds 50 load nodes and supplies 107.8-MVA. The system has 2 substations that can be resized, and there is a possibility of building two more substations. For the reinforcement/construction of the circuits two types of conductors are considered. The minimum voltage magnitude is equal to 0.95pu, the cost of energy losses is 0.1US$/kWh, the loss factor equals 0.35, the interest rate is 0.1, the planning period extends to 20 years and the operation cost of the substations is 0.1US$/kVAh^2.

The optimization process found a solution of US$3328955 in a time of 140 s. This solution is better that the one presented in [11,12], which has a cost of US$3515435. The expansion plan consists of the construction of 2 substations, and all circuits are built using conductor type 1. Additionally, circuit 8-7 was disconnected and circuits 18-17, 22-9, 8-25, 27-8, 28-6, 10-31, 43-13, 33-39, 16-40 and 47-42 were not built. The expansion plan for the 54-node distribution system is shown in Fig. 2.

3.3. Modified 54-node distribution system

The 54-node test system also was used to illustrate the application of the proposed model for the DSEP problem considering allocation of fixed capacitors banks. For this test (shown in Table 1 as mod. 54-node), the operation cost of the substations was disregarded. The solution of the DSEP problem for the modified 54-node distribution system (without capacitor allocation) has an objective function of US$327277, which resizes the substation at node 52 and build the substation at node 53. The solution considering allocation of fixed capacitor banks has an objective function of US$227074, which allocates capacitors in nodes 8, 9 and 43 with an investment cost of US$600 and resizes substations at nodes 51 and 52. This solution is lower than the one mentioned above, mainly because the capacitors make it possible to reduce the loading of the substations, avoiding to build new substations. This solution proposes not to build circuits 22, 23, 24, 32, 33, 38, 45, 50, 51, 52 and 55. This test shows that the DSEP considering allocation of fixed capacitor banks leads to better solutions, reducing real power losses and investment costs in circuits, as it could avoid unnecessary investments in new substations.

3.4. 136-node Distribution system

The 136-node distribution system has a nominal voltage of 13.8-kV, feeds 134 load nodes and supplies 18.118-MVA through two substations with the capacity of 15 MVA and 10 MVA respectively. The minimum voltage magnitude is equal to 0.95 pu, while the maximum over-voltage is 5%. The cost of energy losses is 0.1US$/kWh, the loss factor equals 0.35, the interest rate is 0.1,

Table 2
Results summary for the capacitor allocation tests.

Test	Losses cost (US$)	Investment cost (US$)	Capacitors (kVAr) Node	Capacitors (kVAr) Load level 1	2	3	Power Loses (kW) Load level 1	2	3	Min. Voltage Mag. (pu) Load level 1	2	3	Time (s)
IS-69	72942	–	–	–	–	–	225.03	138.92	51.62	0.9092	0.9288	0.9567	–
CA-69	52510	12600	11	600	600	600	204.81	93.82	36.14	0.9503	0.9503	0.9676	177
			50	1200	900	600							
			53	1200	300	0							
IS-136	201639	–	–	–	–	–	320.37	1117.97	77.01	0.9307	0.8636	0.9669	–
CA1-136	181845	10200	17	600	600	600	287.53	1017.17	69.85	0.9607	0.8977	0.9813	1073
			39	600	600	600							
			155	1200	1200	600							
CA2-136	182191	10200	17	600	600	600	288.55	1015.77	70.09	0.9642	0.9016	0.9830	1202
			39	600	600	600							
			157	1200	1200	600							

Table 3
Results comparison for the test systems (total cost US$).

System	Proposed method	Ref. [11]	Ref. [12]
23-node	170969	172110	170969
54-node	3328955	3515435	3515425
136-node	5457488	5473558	5458927

Table 4
Results comparison for the capacitor allocation tests (total cost US$).

System	Proposed method	Ref. [31]	Ref. [34]
CA-69	65110	66995	–
CA1-136	192045	192046	192074
CA2-136	192388	192389	–

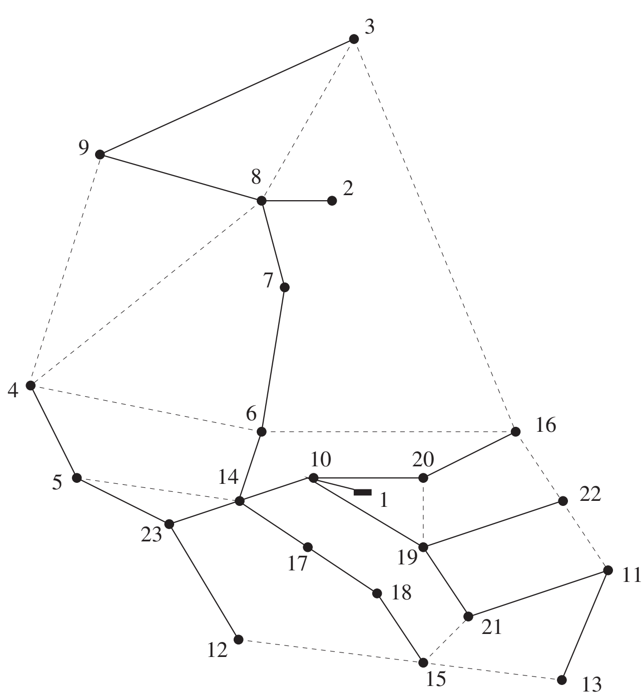

Fig. 1. Expansion plan for the 23-node distribution system.

the planning period extends to 20 years and the operation cost of the substations is 0.1US$/kVAh^2.

The optimization process found a solution of US$5457488 in a time of 1028 s. This solution is better than the one presented in [11] (with a cost of US$5473558) and [12] (with a cost of US$5458927). The expansion plan disconnected the circuits 55, 60, 84, 95, 106, 108, 110 and 134, while the circuits 135, 138, 141, 143, 144, 146, 147, 148 and 149 were built.

3.5. 69-node Distribution system for capacitor allocation

The 69-node distribution system data are available in [27]. It is a 12.66-kV distribution system, feeds 69 load nodes and supplies 4.66-MVA. The switch equipment cost of the switched capacitors is equal to US$300, and the maximum number of fixed and switched capacitors that can be added on to the system is equal to 6. Three load levels are considered, obtained by multiplying of the loads by the factors 1.0, 0.8 and 0.5, with respective durations of 1000, 6760 and 1000 h. The minimum voltage magnitude is equal to 0.95 pu. Table 2 shows a summary of the obtained results (CA-69), including the initial states of the system (IS-69).

Note that the minimum voltage magnitude of system in all load levels are higher than 0.95 pu, and the power losses in all load levels are lower than their initial states. One fixed capacitor and two switched capacitors are allocated, with a total cost of US$65110. These result is better than the solution presented in [31] (with a cost of US$66995).

3.6. 136-node Distribution system for capacitor allocation

The data for the 136-node distribution system are available in [34]. It is a 13.8-kV distribution system, feeds 107 load nodes and supplies 19.96-MVA. Three load levels are considered, obtained by multiplication of the loads by the factors 1.0, 1.8 and 0.5, with respective durations of 6760, 1000 and 1000 h. The switch equipment cost of the switched capacitors is equal to zero, and the maximum number of fixed and switched capacitors that can be added on to the system is equal to 4. For this system two tests were performed: (1) CA1-136 with $\underline{V}$ equal to 0.85 pu (CA1), and (2) CA with $\underline{V}$ equal to 0.90 pu (CA2-136). Table 2 shows a summary of the obtained results, including the initial state of the system (IS-136).

Note that in all tests the system's minimum voltage magnitude in all load levels is higher than the respective value for $\underline{V}$, and the power losses in all load levels are lower than their initial states. In the CA1-136 test, two fixed capacitors and one switched capacitor are allocated, with a total cost of US$192045. This solution is better than the solution presented in [34], at a total cost of US$192074, which was found using Tabu Search. Also, the solution found using the proposed method is better than the solution presented in [31], at a total cost of US$192046.

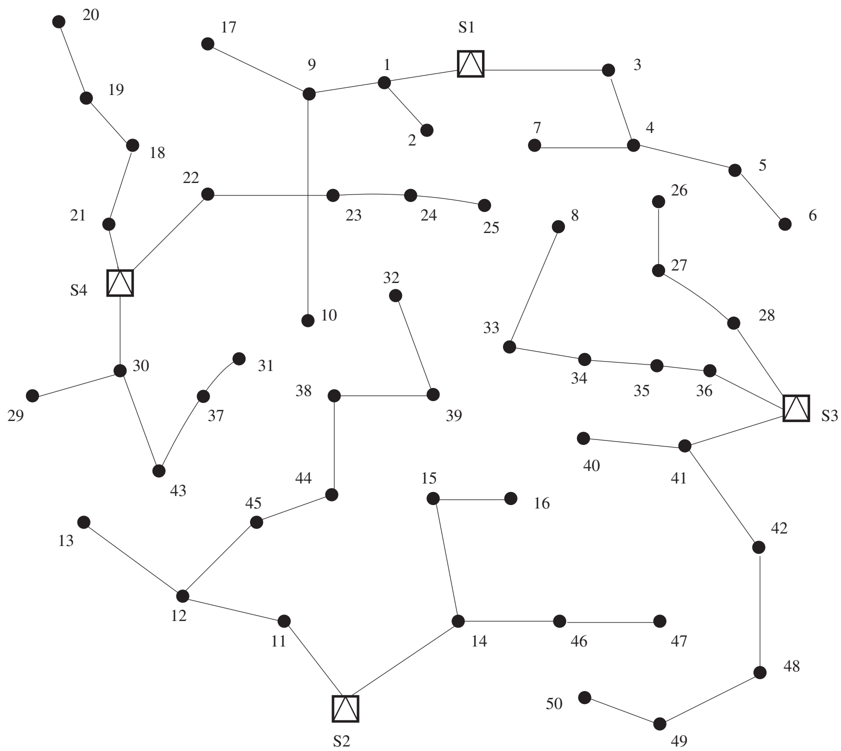

Fig. 2. Expansion plan for the 54-node distribution system.

In the CA2-136 test, two fixed capacitors and one switched capacitor are allocated, with a total cost of US$192388. This solution is better than the solution presented in [31], at a total cost of US$192389. Note that to regard the minimum voltage limit of 0.90 pu in the CA2-136 test, it is necessary to move the capacitor at node 155 in the CA1-136 test to node 157, which causes an increase in power losses.

The time necessary to solve each test, which is shown in the last column of Table 2, is considerably lower in comparison with the times presented in [31], showing the efficiency of the proposed method.

4. Conclusions

A mixed-integer quadratically-constrained programming (MIQCP) model for the distribution system expansion planning (DSEP) problem was presented. The use of a MIQCP formulation has the following benefits: (a) a robust mathematical model that finds the same solution of the MINLP model; (b) an efficient computational behavior with conventional MIQCP solvers; and (c) convergence to optimality is guaranteed using classical optimization techniques. The DSEP model considered the construction/reinforcement of substations, the construction/reconductoring of circuits, the allocation of fixed capacitors banks and the radial topology modification. It is demonstrated that it is possible to obtain the steady-state operation of a radial distribution system using a quadratically-constrained programming problem. That demonstration was used to formulate the DSEP problem as a MIQCP model. Test systems with 23 and 54 nodes and one real distribution systems with 136 nodes were used to show the efficiency of the proposed model for the DSEP problem. The use of the proposed MIQCP model makes it possible to find better solutions than the ones presented in [11,12], with a lesser computational effort. Furthermore, tests carried out for the capacitors banks allocation problem demonstrate the superior performance of the proposed formulation in comparison with the other models [31,34]. We verified that the proposed MIQCP model is efficient and can be considered a useful tool in solving the DSEP problem. Additionally, this novel quadratically-constrained programming model able to obtain the steady-state operation of a radial distribution system can be used to model other optimization problems in electrical distribution systems.

Appendix A

In this section it is demonstrated that the quadratically-constrained programming model (5) represents the steady-state operation of a radial distribution system. It will be demonstrated that, in the solution point of the aforementioned optimization problem, (5).b is active, which implies that the model is equivalent to problem (1)–(3). In order to simplify the formulation, it is supposed that for circuit *ki*, node *k* is closer to the substation than node *i*. This, in addition to the radial operation of the distribution system, means that there is only one circuit *ki* that connects node *i* to the substation. Eq. (36) is the Lagrangian function associated with problem (5), and $\phi_i, \omega_i, \pi_{ij}$ and ρ_{ij} are the dual variables associated with (1), (2), (3) and (5) respectively.

$$\begin{aligned}\mathcal{L} = &\sum_{ij\in\Omega_l} R_{ij}\widehat{I}_{ij}^{sqr} + \sum_{i\in\Omega_b}\phi_i\left(\widehat{P}_{ki} - \sum_{ij\in\Omega_l}\left(\widehat{P}_{ij} + R_{ij}\widehat{I}_{ij}^{sqr}\right) + Pg_i - Pd_i\right)\\ &+ \sum_{i\in\Omega_b}\omega_i\left(\widehat{Q}_{ki} - \sum_{ij\in\Omega_l}\left(\widehat{Q}_{ij} + X_{ij}\widehat{I}_{ij}^{sqr}\right) + Qg_i - Qd_i\right)\\ &+ \sum_{ij\in\Omega_l}\pi_{ij}\left(V_i^{sqr} - V_j^{sqr} - 2(R_{ij}\widehat{P}_{ij} + X_{ij}\widehat{Q}_{ij}) - Z_{ij}^2\widehat{I}_{ij}^{sqr}\right)\\ &+ \sum_{ij\in\Omega_l}\rho_{ij}\left(\widehat{P}_{ij}^2 + \widehat{Q}_{ij}^2 - V_j^{sqr}\widehat{I}_{ij}^{sqr}\right) \qquad (36)\end{aligned}$$

From the Karush–Kuhn–Tucker optimality conditions satisfied by the solution of the optimization problem, we have (37)–(46).

$$\frac{\partial \mathcal{L}}{\partial \widehat{I}_{ij}^{sqr}} = R_{ij} - R_{ij}\phi_i - X_{ij}\omega_{ij} - Z_{ij}^2\pi_{ij} - \rho_{ij}V_j^{sqr} = 0 \quad \forall ij \in \Omega_l \tag{37}$$

$$\frac{\partial \mathcal{L}}{\partial Pg_i} = \phi_i = 0 \quad \forall i \in \Omega_s \tag{38}$$

$$\frac{\partial \mathcal{L}}{\partial Qg_i} = \omega_i = 0 \quad \forall i \in \Omega_s \tag{39}$$

$$\frac{\partial \mathcal{L}}{\partial \widehat{P}_{ij}} = -\phi_i + \phi_j - 2R_{ij}\pi_{ij} + 2\widehat{P}_{ij}\rho_{ij} = 0 \quad \forall ij \in \Omega_l \tag{40}$$

$$\frac{\partial \mathcal{L}}{\partial \widehat{Q}_{ij}} = -\omega_i + \omega_j - 2X_{ij}\pi_{ij} + 2\widehat{Q}_{ij}\rho_{ij} = 0 \quad \forall ij \in \Omega_l \tag{41}$$

$$\frac{\partial \mathcal{L}}{\partial V_i^{sqr}} = \sum_{ij \in \Omega_l} \pi_{ij} - \pi_{ki} - \widehat{I}_{ij}^{sqr}\rho_{ij} = 0 \quad \forall i \in \Omega_b \tag{42}$$

$$\frac{\partial \mathcal{L}}{\partial \phi_i} = \widehat{P}_{ki} - \sum_{ij \in \Omega_l}\left(\widehat{P}_{ij} + R_{ij}\widehat{I}_{ij}^{sqr}\right) + Pg_i - Pd_i = 0 \quad \forall i \in \Omega_b \tag{43}$$

$$\frac{\partial \mathcal{L}}{\partial \omega_i} = \widehat{Q}_{ki} - \sum_{ij \in \Omega_l}\left(\widehat{Q}_{ij} + X_{ij}\widehat{I}_{ij}^{sqr}\right) + Qg_i - Qd_i = 0 \quad \forall i \in \Omega_b \tag{44}$$

$$\frac{\partial \mathcal{L}}{\partial \pi_i} = V_i^{sqr} - V_j^{sqr} - 2(R_{ij}\widehat{P}_{ij} + X_{ij}\widehat{Q}_{ij}) - Z_{ij}^2\widehat{I}_{ij}^{sqr} = 0 \quad \forall i \in \Omega_b \tag{45}$$

$$\frac{\partial \mathcal{L}}{\partial \rho_i} \to \widehat{P}_{ij}^2 + \widehat{Q}_{ij}^2 - V_j^{sqr}\widehat{I}_{ij}^{sqr} = 0 \wedge \rho_{ij} > 0 \vee \widehat{P}_{ij}^2 + \widehat{Q}_{ij}^2 - V_j^{sqr}\widehat{I}_{ij}^{sqr} > 0 \wedge \rho_{ij} = 0 \quad \forall ij \in \Omega_l \tag{46}$$

Constraint (46) represents the complementarity conditions of the optimization problem and depends on the state (active or not) of (5). We will demonstrate that (37)–(46) imply that necessarily $\rho_{ij} > 0$, and thereby (5.b) is activated. Using (42), an expression for π_{ki}, in terms of the variables associated with the circuits in the forward direction of circuit ki, can be written, as shown in (47).

$$\pi_{ki} = \sum_{ij \in \Omega_l} \pi_{ij} - \widehat{I}_{ij}^{sqr}\rho_{ij} \quad \forall i \in \Omega_b \tag{47}$$

As for a terminal node i, no circuits ij exist, therefore $\pi_{ki} = -\widehat{I}_{ij}^{sqr}\rho_{ij}$. So, π_{ki} can be expressed in terms of the variables ρ_{ij} associated with the circuits in the forward direction from node k (set of circuits Ω_{F_k} from node i to the end of the feeder), according to (48).

$$\pi_{ki} = -\sum_{ij \in \Omega_{F_k}} \widehat{I}_{ij}^{sqr}\rho_{ij} \quad \forall i \in \Omega_b \tag{48}$$

From (38) and (39), it is known that ϕ_i and ω_i are zero for the substation node, therefore, using (40) and (41), we can write a recursive relation in order to express ϕ_j and ω_j, as is shown in (49) and (50).

$$\phi_j = \phi_i + 2R_{ij}\pi_{ij} - 2\widehat{P}_{ij}\rho_{ij} \quad \forall ij \in \Omega_l \tag{49}$$

$$\omega_j = \omega_i + 2X_{ij}\pi_{ij} - 2\widehat{Q}_{ij}\rho_{ij} \quad \forall ij \in \Omega_l \tag{50}$$

Using (48), ϕ_j and ω_j are written as functions of the variables associated with the circuits in the backward direction from node j (set of circuits Ω_{B_j} from node j to the substation), according to (51) and (52).

$$\phi_j = -2\sum_{hk \in \Omega_{B_j}}\left[\widehat{P}_{hk}\rho_{hk} + R_{hk}\sum_{mn \in \Omega_{F_h}}\widehat{I}_{mn}^{sqr}\rho_{mn}\right] \quad \forall ij \in \Omega_l \tag{51}$$

$$\omega_j = -2\sum_{hk \in \Omega_{B_j}}\left[\widehat{Q}_{hk}\rho_{hk} + X_{hk}\sum_{mn \in \Omega_{F_h}}\widehat{I}_{mn}^{sqr}\rho_{mn}\right] \quad \forall ij \in \Omega_l \tag{52}$$

Eq. (37) is written as (53).

$$R_{ij} = V_j^{sqr}\rho_{ij} + Z_{ij}^2\pi_{ij} + R_{ij}\phi_i + X_{ij}\omega_{ij} \quad \forall ij \in \Omega_l \tag{53}$$

Using (48), (51) and (52) in (53), we have (54).

$$R_{ij} = V_j^{sqr}\rho_{ij} - Z_{ij}^2\sum_{mn \in \Omega_{F_i}}\widehat{I}_{mn}^{sqr}\rho_{mn} - 2R_{ij}\sum_{hk \in \Omega_{B_i}}\left[\widehat{P}_{hk}\rho_{hk} + R_{hk}\sum_{mn \in \Omega_{F_h}}\widehat{I}_{mn}^{sqr}\rho_{mn}\right] - 2X_{ij}\sum_{hk \in \Omega_{B_i}}\left[\widehat{Q}_{hk}\rho_{hk} + X_{hk}\sum_{mn \in \Omega_{F_h}}\widehat{I}_{mn}^{sqr}\rho_{mn}\right] \quad \forall ij \in \Omega_l \tag{54}$$

Assuming that a solution for the optimization problem (5) exists and that the real and reactive power flows, voltage magnitudes and current flow magnitudes correspond to that solution, (54) represents a system of linear equations for the variables ρ_{ij}. That system is diagonally dominant because factor V_j^{sqr} associated with ρ_{ij} is much greater than the other coefficients.

Furthermore, if the equations of system (54) are sorted in relation to the proximity of the circuits from the substation, it can be shown that if the power flows are positive then the coefficients above the diagonal are negative. So, the system can be reduced to one with a superior triangular matrix, with negative coefficients outside of the diagonal. As the independent terms R_{ij} are positive, the only possible solution to (54) is $\rho_{ij} > 0, \forall ij \in \Omega_l$. Thus, (5.b) is activated in the solution and the optimization model (5) is equivalent to (1)–(4), which represents the steady-state operation of a radial distribution system.

Appendix B

The notation used throughout this paper is reproduced below for quick reference.

Sets:

Ω_a sets of conductor types
Ω_b sets of nodes
Ω_l sets of branches
Ω_s sets of substation nodes

Constants:

α number of hours in one year (8760 h)
λ number of years of the planning period
κ_c capital recovery rate of capacitor constructions
κ_l capital recovery rate of circuit constructions
κ_s capital recovery rate of substation construction/reinforcement
ϕ_l loss factor of circuits
ϕ_s loss factor of substations
τ_l interest rate of real power losses cost
τ_s interest rate of substation operation cost
$\overline{b}$ upper bound for the variable b_{ij}
c^{fx} installation cost of the fixed capacitors banks (US\$)
c_i^s substation fixed cost at node i (US\$)
c_i^v substation operation cost at node i (US\$/(kW)^2/h)
$c_{ij,a}^f$ construction cost of circuit ij using conductor type a (US\$/kWh)
c^{un} annualized cost of each standard capacitor unit (US\$)
c^l energy cost (US\$/kWh)
$\overline{I}_a$ maximum current flow magnitude of conductor type a (A)
l_{ij} length of circuit ij (km)
$\overline{n^{cp}}$ maximum number of capacitors that can be added on to the system
$\overline{n_b^{cp}}$ maximum number of standard capacitor units that can be installed in a node of the system
Sg_i^o maximum apparent power limit of existent substation at node i (kVA)

Sg_i' maximum apparent power limit of substation construction/reinforcement at node i (kVA)
$\underline{V}$ minimum voltage magnitude (kV)
$\overline{V}$ maximum voltage magnitude (kV)
P_i^D real power demand at node i (kW)
Q_i^D reactive power demand at node i (kVAr)
Q^{cp} reactive power of the standard capacitor unit (kVAr)
R_a resistance of conductor type a (Ω/km)
R_{ij} resistance of circuit ij (Ω)
X_a reactance of conductor type a (Ω/km)
X_{ij} reactance of circuit ij (Ω)
Z_a impedance of conductor type a (Ω/km)
Z_{ij} impedance of circuit ij (Ω)

Variables:

$I_{ij,a}^{sqr}$ square of the current flow magnitude of circuit ij associated with conductor type a
$\widehat{I}_{ij}^{sqr}$ square of the current flow magnitude of circuit ij
V_i^{sqr} square of the voltage magnitude at node i
Sg_i^{sqr} square of the apparent power provided by substation at node i
$P_{ij,a}$ real power flow of circuit ij associated with conductor type a
$Q_{ij,a}$ reactive power flow of circuit ij associated with conductor type a
$\widehat{P}_{ij}$ real power flow of circuit ij
$\widehat{Q}_{ij}$ reactive power flow of circuit ij
P_i^S real power provided by substation at node i
Q_i^S reactive power provided by substation at node i
b_{ij} variable used in the calculation of the voltage magnitude drop of circuit ij
n_i^{cp} integer number of standard capacitor units installed at node i
q_i binary variable for allocation of a capacitor at node i
y_{ij}^+ binary variable associated with the forward direction of circuit ij
y_{ij}^- binary variable associated with the backward direction of circuit ij
w_i binary variable for construction/reinforcement of substation at node i
$z_{ij,a}$ binary variable for construction/reconductoring of circuit ij using conductor type a

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